SOFTWARE ENGINEERING LAB PROJECT

HOSPITAL MANAGEMENT SYSTEM

PROJECT REPORT

Submitted by:

Supriya Choudhary 241001001237

Saloni rani 241001001239

Swayam bhalotia 241001001240

Prince kumar 241001001241

Krishnendu dhar 241001001243

1. Introduction

The Hospital Management System (HMS) is a software solution designed to manage and integrate the major administrative and clinical activities of a hospital. The proposed system provides a centralized platform for patient registration, appointment scheduling, doctor and staff management, medical records, billing, notifications, and report generation. The objective is to reduce manual work, improve information availability, minimize duplication of records, and support efficient hospital operations.

This report covers the Initialization and Planning phases of the project, as required for the current Software Engineering laboratory submission. The structure follows the topics specified for the lab: problem identification, objectives, stakeholders, project scope, use case diagram, requirements, resources, schedule, risks, and activity diagram.

2. Project Initialization

2.1 Identification of the Problem

Hospitals handle a large volume of patient, doctor, appointment, treatment, billing, and staff information. When these activities depend heavily on registers, disconnected files, or separate manual processes, common problems include duplicate data entry, difficulty locating patient records, appointment conflicts, delays in billing, and limited visibility of hospital operations.

The proposed HMS addresses these problems by maintaining a centralized system in which authorized users can record, retrieve, update, and manage hospital information according to their roles.

2.2 Objectives of the Project

- To maintain centralized and organized patient information.
- To provide efficient patient registration and profile management.
- To schedule, update, and track doctor appointments.
- To maintain electronic medical records and treatment information.
- To manage doctor, staff, and administrative information.
- To generate accurate bills and maintain payment-related records.
- To provide role-based access so that users can access functions relevant to their responsibilities.
- To generate useful operational reports and notifications.
- To reduce manual effort, data duplication, and avoidable delays in hospital workflows.

2.3 Stakeholders

Stakeholder	Role / Interest	Major Expectations
Patients	Receive healthcare services and appointments	Easy registration, appointments, records and billing information
Doctors	Provide diagnosis and treatment	Quick access to patient history and ability to update medical records
Receptionists	Handle front-desk operations	Fast registration, appointment scheduling and patient lookup
Hospital Administrator	Manage hospital operations	Control users, staff, services, reports and system configuration
Accountant / Billing Staff	Manage financial transactions	Accurate billing, payment records and financial reports
Hospital Management	Monitor overall performance	Reliable reports and timely

		operational information

2.4 Project Scope

In Scope:

- User login and role-based access.
- Patient registration and patient profile management.
- Doctor and staff information management.
- Appointment creation, rescheduling and cancellation.
- Patient medical-record entry and retrieval.
- Billing and payment record management.
- Basic reports and notifications.
- Data storage, retrieval, update and validation.

Out of Scope for the present academic version:

- Advanced AI-based diagnosis or clinical decision making.
- Direct integration with external insurance or government systems.
- Real-time ambulance/GPS tracking.
- Full laboratory equipment integration.
- Large-scale multi-hospital deployment and production-grade regulatory certification.

2.5 Use Case Diagram

The use case diagram identifies the principal actors and the services they interact with in the Hospital Management System. The main actors considered are Patient, Receptionist, Doctor, Administrator, and Accountant.

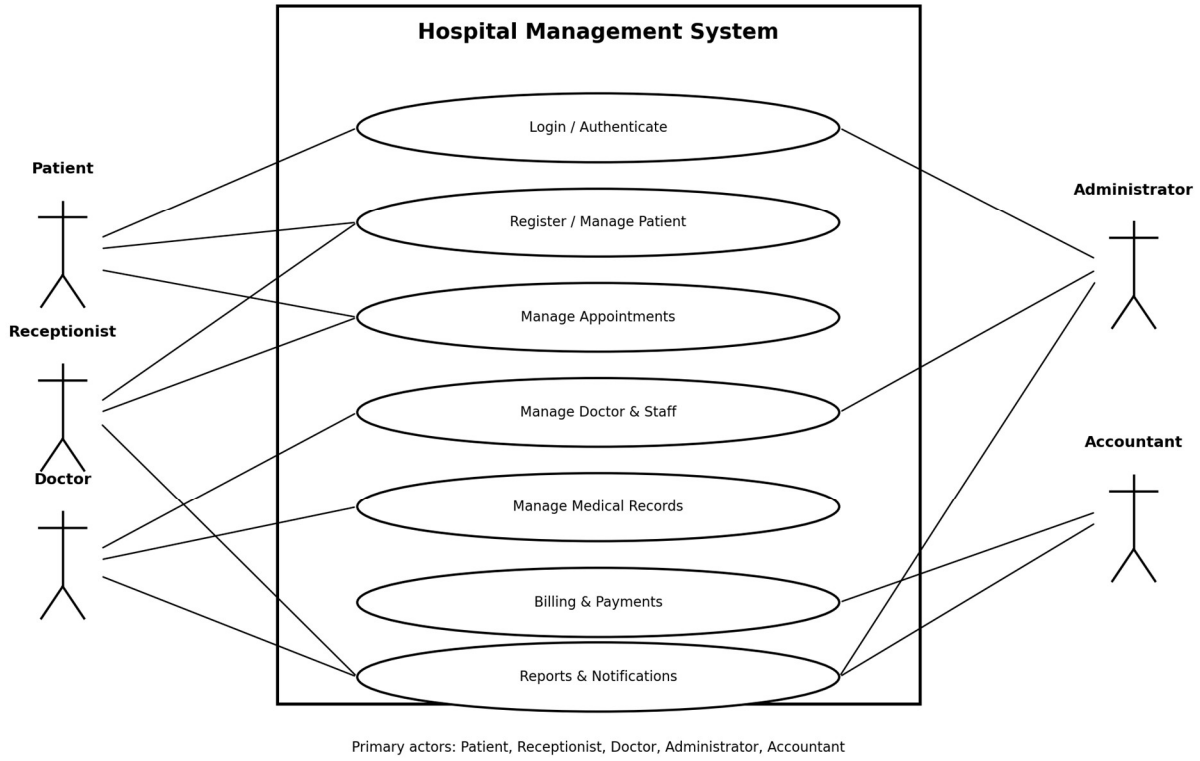


Figure 1: Use Case Diagram of the Hospital Management System

3. Project Planning

Project planning converts the identified problem and objectives into an organized development plan. For this academic project, planning focuses on requirements, resources, schedule, and risks. The planning approach is consistent with the sample material's emphasis on project planning and phase-wise software development documentation.

3.1 Requirements

3.1.1 Functional Requirements

ID	Requirement	Description
FR-01	User Authentication	The system shall allow authorized users to log in and log out securely.
FR-02	Patient Management	The system shall allow authorized staff to register, view, update and search patient information.
FR-03	Doctor & Staff Management	The system shall maintain basic doctor and staff details and their roles.
FR-04	Appointment Management	The system shall allow users to create, view, reschedule and cancel appointments.
FR-05	Medical Records	The system shall allow authorized

		doctors to record and view diagnosis, treatment and related patient information.
FR-06	Billing	The system shall generate and maintain patient bills and payment records.
FR-07	Notifications	The system shall provide appointment or service-related notifications where applicable.
FR-08	Reports	The system shall generate basic reports for hospital administration and billing.
FR-09	Search & Retrieval	The system shall provide search facilities for patients, appointments and records.
FR-10	Access Control	The system shall restrict functions and data according to user roles.

3.1.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Patient and hospital data shall be protected through authentication, authorization and secure data handling.
NFR-02	Performance	Common operations such as login, patient search and appointment retrieval should respond within an acceptable time under normal academic-project load.
NFR-03	Usability	The interface should be simple, consistent and understandable for hospital staff with basic computer knowledge.
NFR-04	Reliability	The system should preserve valid records and handle failures without unnecessary data loss.
NFR-05	Availability	The system should be available during hospital operating hours, subject to infrastructure availability.
NFR-06	Maintainability	The software should use modular components and clear documentation so that future changes are manageable.
NFR-07	Scalability	The design should permit addition of users, departments and records as the system grows.
NFR-08	Data Integrity	The system should validate input and maintain consistent patient, appointment and billing data.

3.2 Resources

3.2.1 Human Resources

- Project Team / Developers – design, coding, database and integration.
- UI/UX or Front-End Member – interface design and usability.
- Database Member – database schema, queries and data integrity.
- Tester / Documentation Member – test preparation, verification and project documentation.
- Project Guide / Faculty – review, feedback and academic supervision.

3.2.2 Software Resources

- Operating system: Windows / Linux / macOS.
- Programming environment: a suitable IDE or code editor.
- Database: MySQL or another relational database suitable for the project.
- Version control: Git or an equivalent system.
- Diagram/documentation tools: draw.io, Microsoft Word, or equivalent.

3.2.3 Hardware Resources

- Computer/laptop for development and testing.
- Adequate storage and RAM for the selected development stack.
- Local network/internet access where required for development and collaboration.

3.3 Project Schedule

A tentative academic-project schedule is given below. Dates can be adjusted according to the faculty's submission calendar.

Phase	Activity	Major Deliverables	Tentative Time
1	Initialization	Problem identification, objectives, stakeholders, scope	September
2	Planning	Functional/non-functional requirements, resources, schedule, risks	End of September
3	System Design	Architecture, database design and detailed UML diagrams	October
4	Development	Implementation of modules and database	End of October
5	Testing	Test cases, verification, defect correction	November
6	Documentation	SRS, final report and presentation preparation	End of November

3.4 Risk Analysis

ID	Risk	Probability	Impact	Mitigation
R1	Incomplete or changing requirements	Medium	High	Review requirements with guide/team and maintain a requirement list.
R2	Schedule delay	Medium	High	Divide work into

				weekly milestones and track progress.
R3	Database/data integrity problems	Medium	High	Use validation, constraints, backups and controlled database changes.
R4	Security/privacy issues	Medium	High	Apply authentication, role-based access and avoid unnecessary exposure of patient data.
R5	Integration difficulties	Medium	Medium	Define module interfaces early and integrate incrementally.
R6	Insufficient testing time	Medium	High	Prepare test cases during development instead of waiting until the end.
R7	Technology/tool learning curve	Medium	Medium	Use familiar technologies and conduct short team learning sessions.

3.5 Activity Diagram

The activity diagram represents a simplified end-to-end patient-service workflow. It begins with user authentication, checks whether the login is valid, allows the user to select a service, and branches according to whether an appointment is required. The flow then proceeds to patient/medical record handling, billing, record saving and notification.

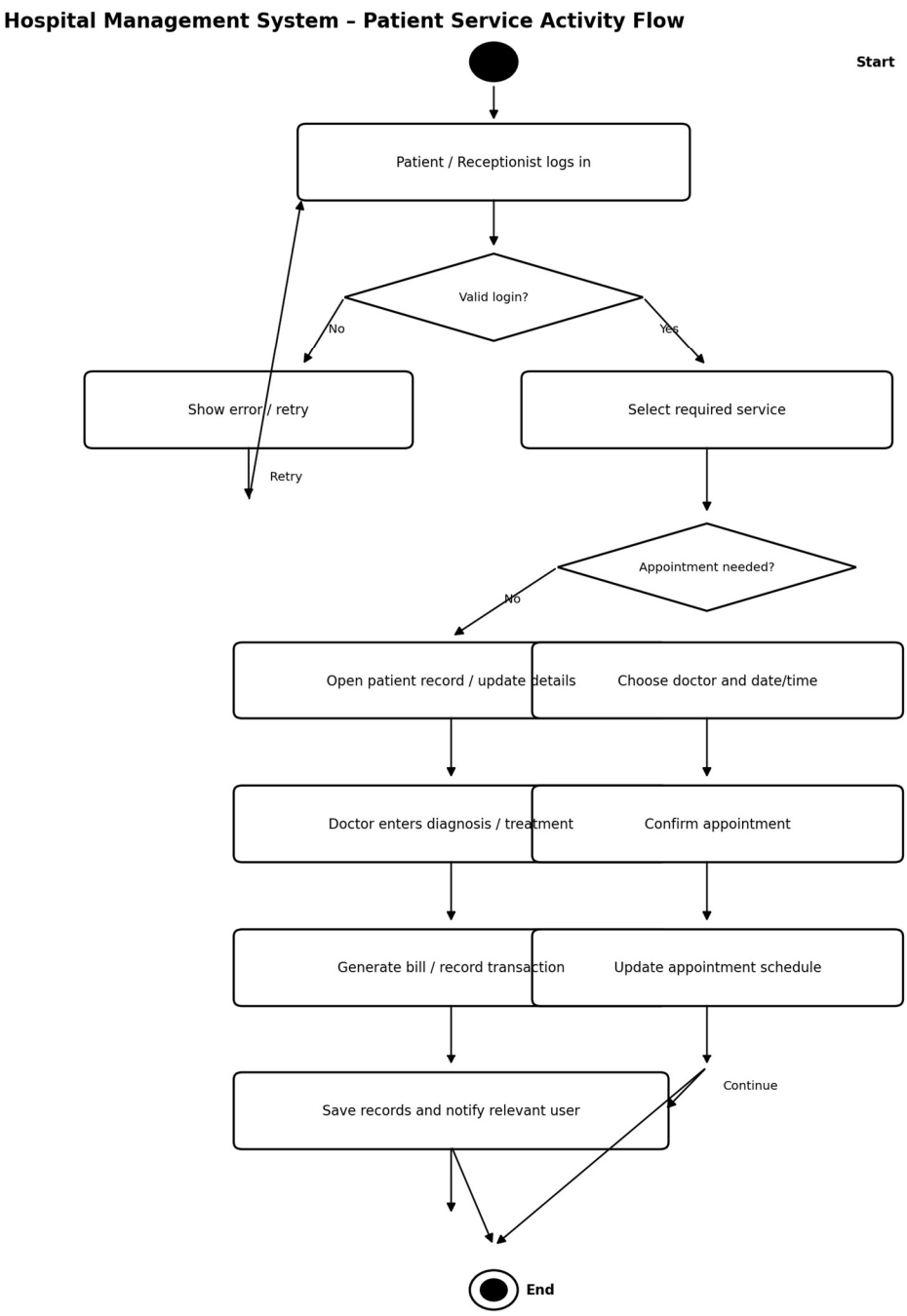


Figure 2: Activity Diagram for a Hospital Management System Service Flow

4. Expected Benefits

- Faster access to patient and appointment information.
- Reduced duplication and manual record maintenance.
- Better coordination between reception, doctors, billing staff and administration.
- Improved traceability of appointments, medical records and billing transactions.
- A structured foundation for later development, testing and maintenance phases.

5. Conclusion

The proposed Hospital Management System aims to provide a centralized and organized approach to hospital information and routine operations. During initialization, the problem, objectives, stakeholders and project scope were identified. During planning, functional and non-functional requirements, resources, schedule and project risks were defined. The use case and activity diagrams provide a visual representation of the system's users and major workflow. These outputs form the baseline for the subsequent system-design, development and testing phases.

6. Reference Basis

The uploaded sample material, “Quality technical education” by Enakshmi Ghosh Kundu, describes phase-wise software development/project-management activities and emphasizes structured project documentation. Its discussion identifies Project Initiation and Project Planning among project management stages. The present report follows the teacher-provided Initialization/Planning structure while adapting the content to the Hospital Management System domain.

Kundu, E. G. (2022). Quality technical education. Tech Vistas, Vol. 3, Issue 2, 43–57.